9.1 Z-Scores, Confidence Intervals, and Margin of Errors

1. Confident Interval for μ (when σ is known)

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z\frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \bar{x} \pm EBM$$

Margin of Error are also called the error bounded for the mean (EBM)

$$EBM = z\frac{\sigma}{\sqrt{n}}$$

The interval is Centered on the Observed Sample Mean. Its margin account for how much sample mean vary from sample to sample.

2. Why this Formula follows from CLT

Recall lesson 8.4 - Central limit Theorem: even when Individual observation are not normally distributed, their sample mean is approximately normal for a sufficiently large sample.

Symbol	Meaning
μ	The Fixed, unknown population mean we want to estimates
σ	The known population Standard Deviation: The spread of individual observations.
n	The number of Observation in one sample
$\bar{X}$	The sample mean as random variable, before observing the sample.
$\bar{x}$	The numerical mean of the Sample we actually Observed.
$SE = \frac{\sigma}{\sqrt{n}}$	The Standard Error: the standard deviation of the sample mean.
z^*	The positive Critical z-values for the choosen confidence level; This is the z in the original formula.

- **Step 1 - Find the Center and Spread of the Sample mean**

Suppose $X_1, \dots, X_n$ are independent observations from the same population, each with mean μ and finite variance $\sigma^2 > 0$

Their sample mean is

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

Taking the expectation gives

$$E(\bar{X}) = \frac{n\mu}{n} = \mu.$$

Independence lets us add the variances. Dividing a random variable by n divide its variances by n^2, so

$$Var(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Taking the square root gives the Standard Deviation

$$SE = SD(\bar{X}) = \sigma/\sqrt{n}$$

These mean and Variance result are exact under the assumption above. The CLT adds the approximately normal shape of the sampling distribution when n is Sufficiently large.

- **Step 2 - Standardize the sample mean**

Substract its mean and divide by its standard deviation:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \text{ is approximately standard normal.}$$

For one individual observation, the denominator would be σ. Here we are studying an **average**, so the denominator is the smaller quantity $\sigma/\sqrt{n}$

- **Step 3 - Select the Central Probability**

Let the confidence level be $1-\alpha$. Choose z^* so that the standard normal curve has central area $1-\alpha$, leaving $\frac{\alpha}{2}$ in each tail:

$$P(-z^* \leq Z \leq z^*) \approx 1-\alpha.$$

for 95% Confidence, $\alpha = 0.05$, so each tail has area 0.025. The left area at the positive cutoff is 0.975 giving $z^* \approx 1.96$. Substitute the standarized sample mean.

$$P\left(-z^* \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z^*\right) \approx 1-\alpha$$

- **Step 4 - Rearrange to put the unknown mean in the middle**

Multiply the inequalities inside the probability statement by the positive standard error:

$SE = \sigma/\sqrt{n}$, so basically $-z^* SE \leq \bar{X} - \mu \leq z^* SE$.

Solve for each side for μ:

$$\bar{X} - \mu \leq z^* SE \rightarrow \mu \geq \bar{X} - z^* SE.$$

$$\bar{X} - \mu \geq -z^* SE \rightarrow \mu \leq \bar{X} + z^* SE.$$

The key switch: Initially, we ask how far the random sample mean can be form fixed population mean. After rearranging, we construct random interval endpoints around the sample mean and ask whether they contain that same fixed population mean.

- **Step 5 - Insert the observed Sample mean**

Once we collect the sample, replace $\bar{X}$ with its observed value $\bar{x}$:

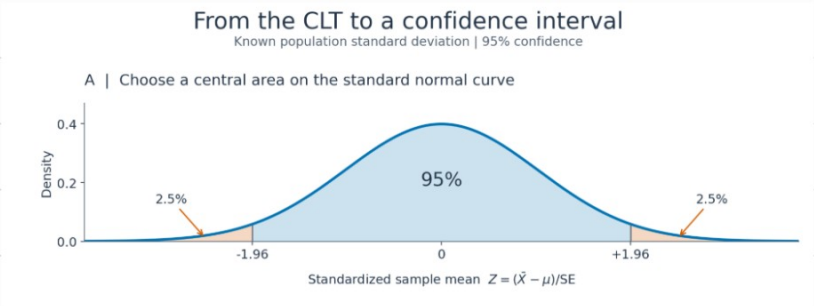
$$CI = \left[\bar{x} - z^* \frac{\sigma}{\sqrt{n}},\ \bar{x} + z^* \frac{\sigma}{\sqrt{n}}\right]$$

The distance from the center to either endpoints is the margin of error:

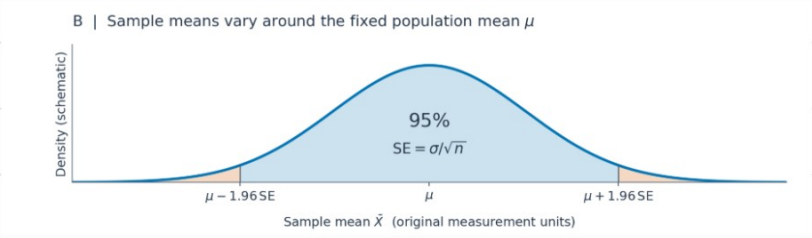
$$EBM = z^* \times SE = z^* \frac{\sigma}{\sqrt{n}}$$

The confidence interval estimates **one population mean** It does not describe where 95% of individual observation lie.

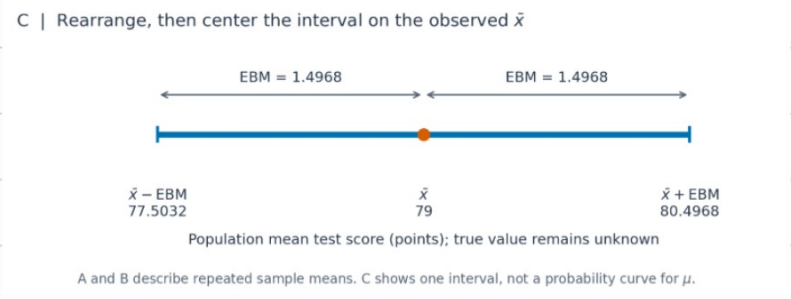
- Illustration - From the normal curve to an interval
  - A: Standardize scale. About 95% of the standard normal area is between -1.96 and +1.96.



  - B: Sampling distribution. Returning to the original units, about 95% of sample means fall within 1.96 SE of the fixed μ. The center is μ, not the observed $\bar{x}$.



  - C: Observed interval. Rearranging gives interval centered on $\bar{x}$. Using example 1 below $SE = 5.4/\sqrt{50} \approx 0.76368$ and $EBM \approx 1.4968$, so the interval is $[77.5032, 80.4968]$. The true μ remains unknown; the figure does not assume that is equals 79.



Panel C is an interval on number line, not a probability distribution for μ

- When this derivation applies
  - The observations should be an independent random sample from the same population, with finite variance. When sampling without replacement from a finite population, independence is an approximation; the usual check is that the sample is no more than about 10% of the population.
  - The population standard deviation σ must be known. If it is estimated by the sample standard deviation S, the usual method is called a t Confidence interval.
  - If the population is normal, the sampling distribution of $\bar{X}$ is exactly normal for any n, and the interval has exact coverage using the exact normal critical value.
  - Otherwise, the interval relies on the CLT approximation $n \geq 30$ is only a rule of thumb: strong skewness or heavy tails can require a much larger sample.

3. Finding the Critical Z-Value

The area to the **left** of the positive critical value z is

$$\text{left area} = \frac{CL+1}{2}$$

Where CL is the confidence level as a decimal. Look this area up in a z-table that gives **left** areas.

Confidence level	Left area	z
90%	0.95	1.645
95%	0.975	1.96
99%	0.995	2.576

For 90%, the table has 0.94950 at $z = 1.64$ and 0.95053 at $z = 1.65$. 0.95 is about halfway between them so average: $z = 1.645$.

4. Example 1 - Physics test, 95% CI

$\sigma = 5.4$, $n = 50$, $\bar{x} = 79$ find the margin of error and its interval

1. z: left area $= \frac{(0.95+1)}{2} = 0.975 \rightarrow z = 1.96$
2. Standard error: $\frac{\sigma}{\sqrt{n}} = 0.76368$
3. (b) Margin of error: $EBM = z \times SE = z \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{5.4}{\sqrt{50}} = 1.4968$
4. (a) Interval: $\bar{x} \pm EBM = \bar{x} + z \times \frac{\sigma}{\sqrt{n}} = 79 \pm 1.4968$

$$77.5032 < \mu < 80.4968$$

We are 95% confident that the mean score of the whole class lies between about 77.50 and 80.50

5. Example 2 - Chemistry test 90% CI

$\sigma = 6.5$, $n = 100$, $\bar{x} = 82$ find the margin of error and its interval.

1. z: left area $= \frac{(0.9+1)}{2} = 0.95 \rightarrow z = 1.645$
2. (b) Margin of error: $EBM = z \times \frac{\sigma}{\sqrt{n}} = 1.645 \times \frac{6.5}{\sqrt{100}} = 1.06925$
3. (a) Interval: 82 ± 1.06925

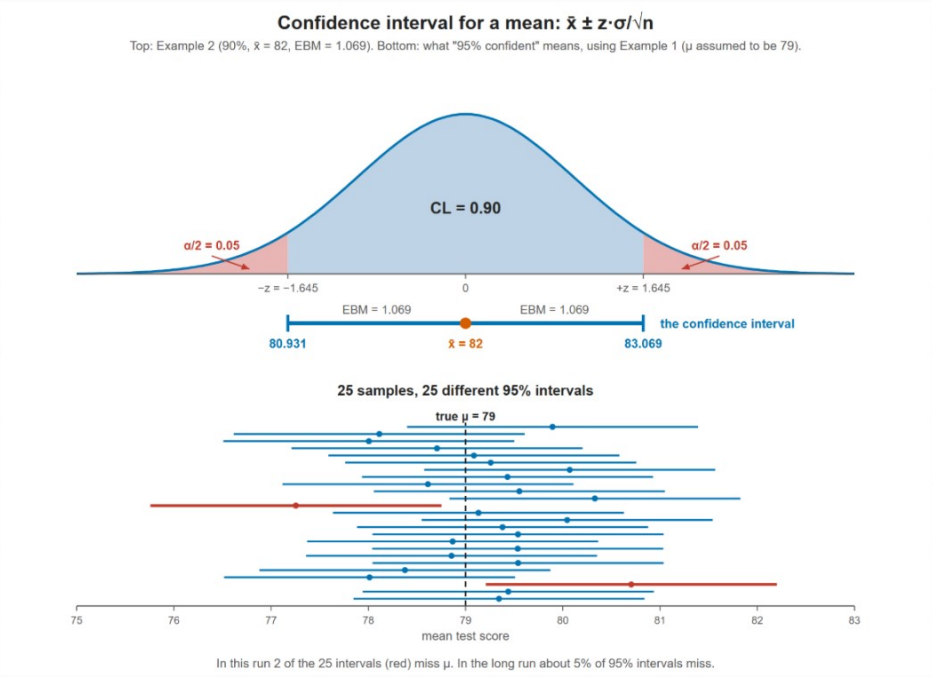
$$80.93075 < \mu < 83.06925$$

6. Confidence Interval vs Confidence level

Term	Meaning	Example
Confidence Interval (CI)	The range of values $(\bar{x} - EBM, \bar{x} + EBM)$	80.93 to 8307
Confidence Level (CL)	The target long-run proportion of intervals covering μ; the central standard normal area between $-z^*$ and $+z^*$ determines the cutoff	0.90
α	The total standard normal tail area $\alpha = 1 - CL$ split as $\alpha/2$ inch tail	0.10 → 0.05 per tail

· Interpreting "95% Confidence":

If we took many samples and built interval from each one, about 95% of those interval would contain the true mean μ, and about 5% would miss it. The video phrases this in term of sample means landing in the interval; this is more precise statement.



- Top: the confidence level is the middle area between $-z$ and $+z$, and the interval reaches one EBM either side of $\bar{x}$.
- Bottom: 25 simulated sample give 25 different intervals. Most of them cover the true mean (dashed line), but a few miss it, which is what "95% confident" means in the long run.

7. What Changes the Margin of Error?

$$EBM = z\frac{\sigma}{\sqrt{n}}$$

Change	Effect on EBM	Why
Higher confidence level (90% → 95%)	Larger	z grows, so the interval gets wider.
Larger sample size n	Smaller	$\sqrt{n}$ is the denominator.

8. Key Takeaways

- $CI = \bar{x} \pm z\frac{\sigma}{\sqrt{n}}$, and $EBM = z\frac{\sigma}{\sqrt{n}}$
- Critical z: look up the left area $\frac{(CL+1)}{2}$. 90% → $z = 1.645$, 95% → $z = 1.96$
- Higher confidence gives a wider interval. A bigger sample gives a narrower one.

9.2 How to Find the Z-Score Given the Confidence Intervals

1. Common Values to Memorize

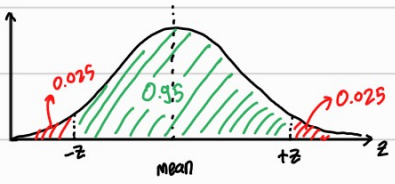
Confidence level	z
90%	1.645
95%	1.96
98%	2.33
99%	2.575

2. Method for Any Confidence Level

- Step 1 - area of the left of $+z$:

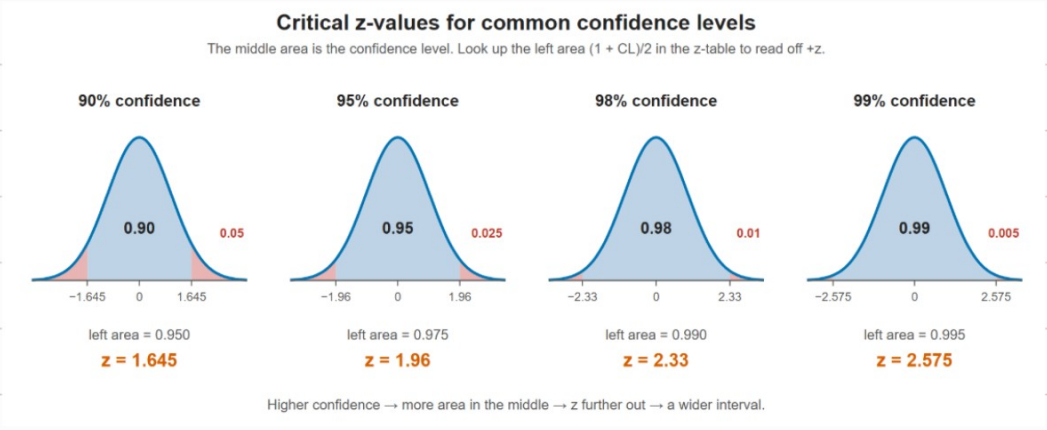
$$\text{left area} = \frac{1+CL}{2}$$

Why: the confidence level is the area between $-z$ and $+z$. The leftover $1-CL$ is split evenly between two tails. For 95%.



- Step 2 - Find that area inside a positive z-table (one that gives left areas). Read the row (ones and tenths digits), then add them.

Example: 0.975 sits in row 1.9 column 0.06 so $z = 1.9 + 0.06 = 1.96$



The confidence level is the middle area, and each tail gets half of what is left. The left area up to $+z$ is $(1+CL)/2$ which is the number to look up in the z-table.

3. Practice

- 97% Confidence

  Left area = $\frac{1+0.97}{2} = 0.985$ → row 2.1, column 0.07 → $z = 2.17$

- 92% Confidence

  Left area = $\frac{1+0.92}{2} = 0.96$

  The closest table entry is 0.9599, in row 1.7, column 0.05 so $z = 1.75$

4. Key Takeaways

- Left area = $(1+CL)/2$. Look it up in the body of the z-table to read off z.
- If the exact area isn't in the table, use closest entry, or average two entries that are equally close (eg. 90% → 1.645).
- Know 1.645, 1.96, 2.33 and 2.575 by heart.

9.3 Student's T-Distributions

1. When to use t instead of z

Use the normal (z) distribution when...	Use Student's t when...
the population SD σ is known	Only the sample SD s is known
$n \geq 30$	$n < 30$

2. Formulas

$$\bar{x} \pm t_{v,\alpha/2}\frac{s}{\sqrt{n}}, \quad EBM = t_{v,\alpha/2}\frac{s}{\sqrt{n}}$$

Symbol	Meaning
$v = n-1$	Degree of freedom (Greek "nu")
$\alpha = 1 - CL$	The total area outside the interval, i.e. both tails
$\alpha/2 = \frac{1-CL}{2}$	The one-tail area, used to look up t

Reading a t-table: go to the row for v and the column for one-tail area $\alpha/2$

3. Example 1 - Student Weights, 95% CI

20 Students: $\bar{x} = 165$ lb, $s = 4.5$ lb

s is a sample SD (the school has far more than 20 students) and $n = 20 < 30$, so use t.

- $v = n - 1 = 20 - 1 = 19$
- $\alpha/2 = \frac{1-CL}{2} = \frac{1-0.95}{2} = 0.025$
- Table: $t_{v,\alpha/2} = t_{19,0.025} = 2.093$

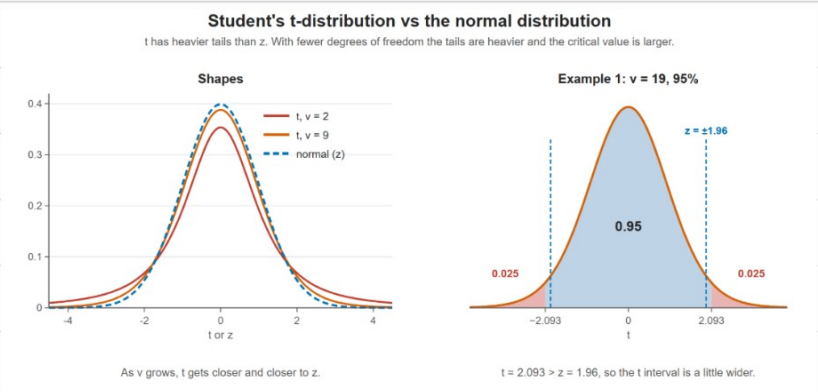
(b) Margin of error

$$EBM = t_{19,0.025} \times \frac{s}{\sqrt{n}} = 2.093 \times \frac{4.5}{\sqrt{20}} = 2.106$$

(a) Interval

$$\bar{x} \pm EBM \rightarrow 165 \pm 2.106 \rightarrow 162.894 < \mu < 167.106$$

We are 95% confident that the mean weight of all school's students is between about 162.9 and 167.1 lb



Left: t has heavier tails than the normal curve, especially with few degrees of freedom.

Right: for $\nu = 19$ the 95% critical value is 2.093, a bit further out than $z = 1.96$ so t intervals are slightly wider.

4. Example 2 - Chemistry Score, 90% CI

The class has 500 students. 10 scores were samples 76, 84, 92, 58, 89, 73, 97, 85, 77.

(a) Sample mean and SD

$$\bar{x} = \frac{\sum x_i}{n} = \frac{800}{10} = 80$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{1234}{9}} \approx 11.709$$

(b) Margin of error at 90%

- $\nu = n - 1 = 10 - 1 = 9$
- $\alpha/2 = \frac{(1 - CL)}{2} = \frac{1 - 0.9}{2} = 0.05$
- Table: $t_{9, 0.05} = 1.8331$

$$EBM = t_{\nu, \alpha/2} \times \frac{s}{\sqrt{n}} = 1.8331 \times \frac{11.709}{\sqrt{10}} \approx 6.787$$

(c) 90% Confidence Interval

$$\bar{x} \pm EBM \rightarrow 73.213 < \mu < 86.787$$

5. Key Takeaways

- Use t when σ is unknown (you only have s), especially when $n < 30$.
- Degrees of Freedom $\nu = n - 1$. Look up the one-tail area $\frac{\alpha}{2} = \frac{(1 - CL)}{2}$
- $CI = \bar{x} \pm t \frac{s}{\sqrt{n}}$ The t value is larger than z for small samples, which gives a wider and more cautious interval.

9.4 Finding a Confidence Interval of a Population Proportion

1. The Big Idea

You want to know what **fraction** of a large group has some property, for example "What percent of student at this university own a laptop?"

Asking everyone (maybe 10000 people) is too much work, so you ask a **sample** instead.

The sample gives you **one estimate**. Because its only a sample, that estimate is probably a little off. A confidence interval turns the single estimate into a **range** and says how confident you are that the true value is inside it:

$$\underbrace{\hat{p}}_{\text{Your Estimate}} \pm \underbrace{E}_{\text{Margin of error}}$$

Earlier lesson did this for a **mean** (an average numbers). This lesson does the same thing for a **proportion** (a percentage of yes answers).

2. The Symbols

The letters are the most confusing part of this lesson, so here they all are in one place:

Symbols	Name	Meaning	Laptop example
n	Sample Size	How many people you asked	800
x	Number of Successes	How many said yes	584
p	Population proportion	The true fraction of everyone. Unknown, and it's what we're trying to estimate	?
$\hat{p}$ ("p-hat") also written as p' ("p-prime")	Sample Proportion	the fraction of your sample that said yes: $\hat{p} = x/n$	$\hat{p} = \frac{x}{n} = \frac{584}{800} = 0.73$
$\hat{q}$ ("q-hat")	Proportion of Failures	the fraction of your sample that said no $\hat{q} = 1 - \hat{p}$	0.27
CL	Confidence level	how sure you want to be, as a decimal	95% → 0.95
z	Critical Value	how many standard errors wide the interval is; depends only on CL	1.96
E or EBP	Margin of error, or "Error bound for a proportion".	how far the interval reaches on each side of $\hat{p}$	0.030

> p vs $\hat{p}$: p (no hat) is the true answer for the whole population, which you never actually know. $\hat{p}$ (with hat) is what your sample gave you. The confidence interval is a range of believable values for p, built around $\hat{p}$.

$\hat{p}$ and $\hat{q}$ always add up to 1. If 73% said yes, then 27% said no

3. Formulas

Quantity	Formula
Sample proportion	$\hat{p} = \frac{x}{n}$
Proportion of Failures	$\hat{q} = 1 - \hat{p}$
Check: is the normal distribution OK?	$n\hat{p}\hat{q} \geq 10$
Area to the left of z	$\frac{1+CL}{2}$
Margin of error	$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$
Confidence Interval	$\hat{p} - E < p < \hat{p} + E$
Sample size needed	$n = \frac{z^2 \hat{p}\hat{q}}{E^2}$, always rounded up

Common z values:

Confidence Level	Area to the left:	z
90%	0.95	1.645
95%	0.975	1.96
98%	0.99	2.33
99%	0.995	2.575

4. The Recipe (do this steps in order)

- 1. write down n and x from the problem
- 2. Find $\hat{p} = x/n$ and $\hat{q} = 1 - \hat{p}$.
- 3. Check $n\hat{p}\hat{q} \geq 10$. If it is, the normal distribution is a good approximation and you can use z.
- 4. Find z from the confidence level (see the next session)
- 5. Find the margin of error $E = z\sqrt{\hat{p}\hat{q}/n}$
- 6. Build the Interval: lower end $\hat{p} - E$, upper end $\hat{p} + E$
- 7. Write a Sentence: Saying what it means.

5. Step 3 in detail - Why check $n\hat{p}\hat{q} \geq 10$?

Each person gives one of two answer (yes or no), So the number of yeses follows a binomial distribution. The confidence interval formula uses the normal (bell) curve instead, because z values come from the normal curve.

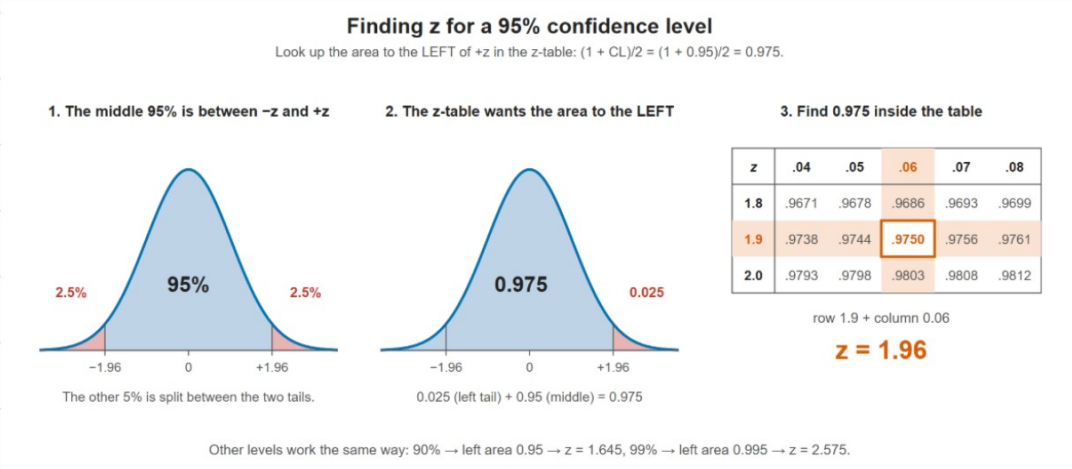
That's only allowed when the binomial distribution looks enough like bell curve, which happens when the sample is big enough and $\hat{p}$ isn't too close to 0 or 1.

the rule of thumb $n\hat{p}\hat{q} \geq 10$ test this.

> Many textbooks use a slightly different check: $n\hat{p} \geq 10$ and $n\hat{q} \geq 10$, meaning at least 10 yesses and 10 noes. Use whichever your course uses.

6. Step 4 in detail - Finding z from the confidence level

The Picture. A 95% confidence level means the middle 95% of the bell curve, between $-z$ and $+z$. The other 5% is split equally between the two tails: 2.5% on the left and 2.5% on the right.



The three steps for 95%:

- **Left:** the confidence level is the middle area, and the leftover 5% is split into two 2.5% tails.
- **Middle:** the z-table measures from the far left, so the area it wants also includes the left tail, 0.975 in total.
- **Right:** 0.9750 sits in row 1.9 and column 0.06, so $z = 1.96$.

A z-table gives the area to the **left** of a z-score. The area to the left of $+z$ is the middle 95% plus the left tail 2.5%:

$$0.95 + 0.025 = 0.975$$

The Shortcut formula gives the same answer without the picture:

$$\text{area to the left} = \frac{1+CL}{2} = \frac{1+0.95}{2} = \frac{1.95}{2} = 0.975$$

Looking it up in the table. Find 0.975 **inside** the positive z-table, then read off its row and column:

- Row: 1.9
- Column: 0.06
- So $z = 1.9 + 0.06 = 1.96$

If the exact area isn't in the table, take the two closest entries and average their z-scores. For example, for 90% the area to the left is 0.95. The table has 0.9495 at $z = 1.64$ and 0.95 at $z = 1.65$

$$z = \frac{1.64 + 1.65}{2} = 1.645$$

4. Step 5 in detail - What the margin of error formula means

$$E = z \times \underbrace{\sqrt{\frac{\hat{p}\hat{q}}{n}}}_{\text{Standard Error}}$$

- The square-root part is the **standard error:** roughly how much $\hat{p}$ would vary if you repeated the survey with different random people.
- z sets how many standard errors you allow on each side.

5. Example 1 - Laptop Ownership

800 randomly selected college students at university XYZ were asked if they own a laptop. 584 said yes.

(a) Can the binomial distribution be approximated by a normal distribution?

-1. Write down n and x from the problem.

n = Sample size = 800

x = Number of successes = 584

-2. Find $\hat{p} = \frac{x}{n}$ and $\hat{q} = 1 - \hat{p}$.

$$\hat{p} = \frac{584}{800} = 0.73$$

-3. Check $n\hat{p}\hat{q} \geq 10$

$n\hat{p}\hat{q} = 800 \times 0.73 \times (1-0.73) = 157 > 10$ thus the normal distribution is a good approximation and you can use z.

(b) 95% Confidence interval and margin of error

-4. Find z from the confidence level

$CL = 95\%$

thus left area $= \frac{1+CL}{2} = \frac{1.95}{2} = 0.975$

thus from table $z = 1.96$

- 5. Find the margin of error

$$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}} = 1.96\sqrt{\frac{0.73 \times (1-0.73)}{800}} = 0.03076$$

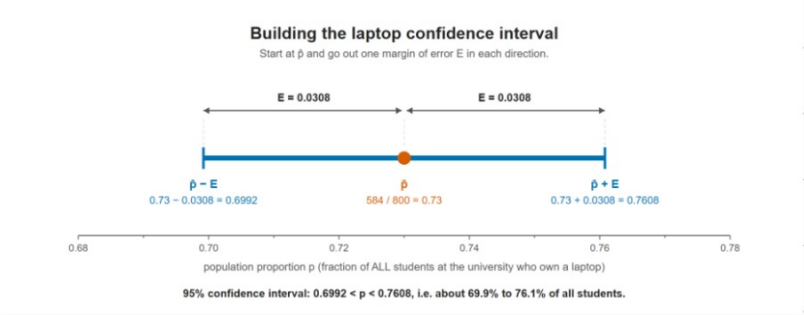
- 6. Build the interval

lower end $= \hat{p} - E = 0.73 - 0.03076 = 0.69924$

higher end $= \hat{p} + E = 0.73 + 0.03076 = 0.76076$

$$\hat{p} - E < p < \hat{p} + E$$

$$0.6992 < p < 0.7608$$



-7 Interpretation:

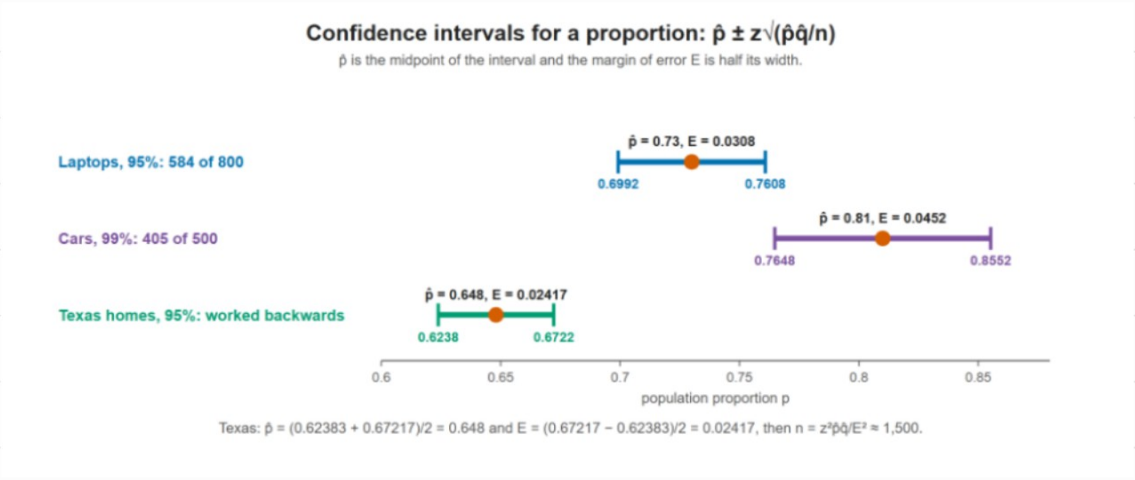
"We are 95% confident that between about 69.9% and 76.1% of all students at university XYZ own a laptop."

The margin of error is 0.0308, about ±3.1 percentage points.

(c) What sample size gives a 2% margin of error at 90% confidence?

- This time we're **not** building an interval. We're asking how many people we would need to survey.
- **What changes.** $\hat{p}=0.73$ and $\hat{q}=0.27$ stay the same. The margin of error is now $E=2\%=0.02$, the confidence level is 90% and n is the unknown.
- We know that $E=z\sqrt{\frac{\hat{p}\hat{q}}{n}} \rightarrow E^2=z^2\frac{\hat{p}\hat{q}}{n} \rightarrow n=\frac{z^2\hat{p}\hat{q}}{E^2}$
- for $CL=90\% \rightarrow$ left area $=1.9 \rightarrow z=2.575$
- Thus $n=z^2\times\frac{\hat{p}\hat{q}}{E^2}=1.645^2\times\frac{0.73\times0.27}{0.02^2}=1333.39$ round up $n=1334$ students

Why always round up? With 1333 student the margin of error would be slightly more than 2%. We need **at least** enough people to reach 2%, so we go up to the next whole person.



6. Example 2 - Car Ownership

500 residents in town ABC were asked if they own a car. 405 said yes.

Write down what you know and find $\hat{p}, \hat{q}$.

$n=500$, $x=405$, $\hat{p}=\frac{x}{n}=\frac{405}{500}=0.81$, $\hat{q}=1-\hat{p}=0.19$

(a) 99% Confidence Interval

Find z. The area to the left is $(1+0.99)/2=0.995$

In the table 0.995 falls between $z=2.57$ and 2.58 so average them:

$$z=\frac{2.57+2.58}{2}=2.575$$

Find the margin of error.

$$E=z\times\sqrt{\frac{\hat{p}\hat{q}}{n}}=2.575\times\sqrt{\frac{0.81\times0.19}{500}}=0.0452$$

Build the interval

$$\hat{p}-E<p<\hat{p}+E \rightarrow 0.81-0.0452<p<0.81+0.0452 \rightarrow \boxed{0.7648<p<0.8552}$$

Interpretation we are 99% confident that between about 76.5% and 85.5% of all residents of town ABC own a car.

(b) What sample size gives a 3% margin of error at 98% confidence?

$$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}} \rightarrow n = z^2 \times \frac{\hat{p}\hat{q}}{E^2}$$

when $CL = 98\% \rightarrow$ left area $= \frac{1 \pm 0.98}{2} \rightarrow z = 2.33$

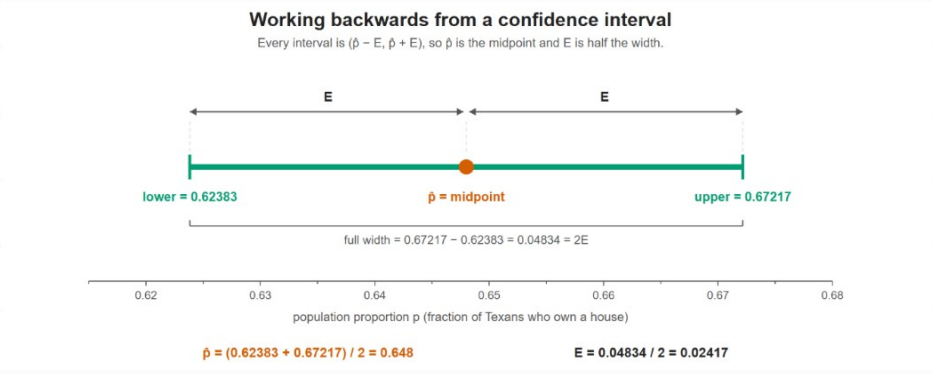
thus $n = 2.33^2 \times \frac{0.81 \times 0.19}{(3\%)^2} = 928.3$ round up $n = 929$

7. Example 3 - Working backwards from an Interval

Some number of people in texas were asked if they own a house. The 95% Confidence Interval for the true proportion of homeowners was (0.62383, 0.67217)

This time the interval is given, and we work back to find the pieces that build it. The key fact is that every confidence interval has the shape $(\hat{p} - E, \hat{p} + E)$

So $\hat{p}$ sits exactly in the middle and E is the distance from the middle to either end.



Only the two ends are given. Because the interval is $\hat{p} \pm E$, $\hat{p}$ is the point halfway between them (0.648) and E is half of the full width (0.04834/2 = 0.02417).

(a) What is the sample proportion?

$\hat{p}$ is the midpoint, the average of the two ends?

$$\hat{p} = \frac{\text{lower} + \text{upper}}{2} = \frac{0.62383 + 0.67217}{2} = \frac{1.296}{2} = 0.648$$

(b) What is the error bound?

Method 1: Upper end minus the middle

$$E = 0.67217 - 0.648 = 0.02417$$

Method 2: half the width of the interval. The full width is $2E$, so

$$E = \frac{\text{upper} - \text{lower}}{2} = \frac{0.67217 - 0.62383}{2} = \frac{0.04834}{2} = 0.02417$$

Both methods give the same answer.

(c) How many people were surveyed?

That's n, so use the sample-size formula. at 95% Confidence $z = 1.96$ and $\hat{q} = 1 - 0.648 = 0.352$:

$$n = \frac{z^2 \hat{p}\hat{q}}{E^2} = \frac{3.8416 \times 0.228096}{0.000584189} = 1499.95$$

So about 1500 people were Surveyed.

(b) How many said yes?

$\hat{p}$ is the fraction who said yes, and x is the number who said yes, Since $\hat{p} = x/n$, multiply both sides by n:

$$x = n \times \hat{p} = 1500 \times 0.648 = 972$$

So 972 people said yes, which is 64.8% of 1500.

8. Common Mistakes

- Mixing up p and $\hat{p}$. $\hat{p}$ comes from the sample, and you calculate it. p is the true population value, and you never know it exactly. The interval is a guess for p.
- Forgetting to convert percentages. A 2% margin of error is $E = 0.02$, not 2. A 95% confidence level is 0.95.
- Looking up the confidence level directly in the z-table. Look up the area to the left, $(1+CL)/2$. For 95%, look up 0.975 not 0.95.
- Forgetting to square z or E in the sample-size formula.
- Rounding n down. Sample size always round up.
- Misreading what "95% confident" means. It does not mean there's 95% chance that p is in that particular interval). p is a fixed number, and it either is or isn't inside. It means the method works 95% of the time: If you repeated the survey many times and built an interval each time, about 95% of those intervals would contain the true p.

9. Key Takeaways

- $\hat{p} = \frac{x}{n}$ is your estimate, and $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal curve.
- Find z by looking up the area to the left $\frac{1+CL}{2}$ in the z table.
- Confidence interval $\hat{p} \pm z\sqrt{\frac{\hat{p}\hat{q}}{n}}$
- Higher confidence means a bigger z and a wider interval. A bigger sample means a narrower interval.
- Given an interval: its midpoint is $\hat{p}$, and half its width is E.
- Sample size: $n = z^2\frac{\hat{p}\hat{q}}{E^2}$, always rounded up. Lesson 9.5 covers sample size in more depth.

9.5 How to Calculate the Sample Size

1. Formulas

For a mean. Start from margin of error:

$$E = z\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \rightarrow n = \frac{z^2\sigma^2}{E^2}$$

For a proportion. Same algebra, starting from the proportion's margin of error (lesson 9.4):

$$E = z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \rightarrow n = \frac{z^2\hat{p}(1-\hat{p})}{E^2}$$

It is the mean formula with σ^2 replaced by $\hat{p}(1-\hat{p})$, as explained below.

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ maximum, so the resulting n is big enough whatever the true proportion is.

Always **round n up** to the next whole number.

2. The logic behind the formulas

1. Every sample gives a different answer. Survey n people and you get a sample proportion $\hat{p}$. A different random group would give a slightly different $\hat{p}$. Across many samples, the $\hat{p}$ values form a bell curve centred on the true p. The spread of that curve is the **standard error**:

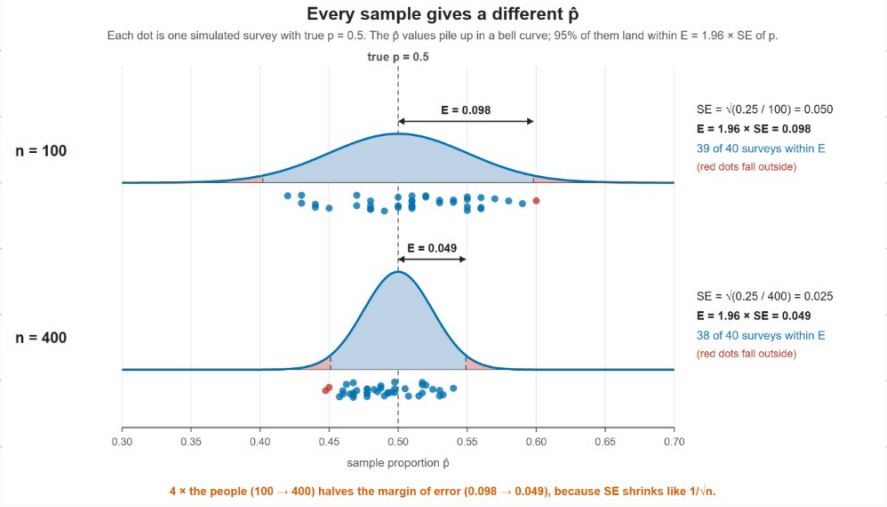
$$SE = \sqrt{\frac{p(1-p)}{n}}$$

Why p here and not $\hat{p}$? The SE describes how much $\hat{p}$ wobbles from sample to sample. That wobble comes from the population the sample are drawn from, so the exact formula uses the true p. But p is exactly what we don't know. In practice we plug in the best stand-in available: the sample's $\hat{p}$ when building an interval from data (lesson 9.4), or a guess, or 0.5, when planning a survey before any data exists (point 6). That swap is why the formulas at the top are written with $\hat{p}$. With a decent-sized sample $\hat{p}$ is close to p. So the estimated SE is close to the true one.

2. The margin of error is z standard error. On a bell curve, 95% of values lie within 1.96 standard deviations of the centre. So 95% of the time, $\hat{p}$ lands within $1.96 \times SE$ of the truth:

$$E = z \times SE$$

E is how far off you are willing to be, and z is how sure you want to be of staying within that distance.



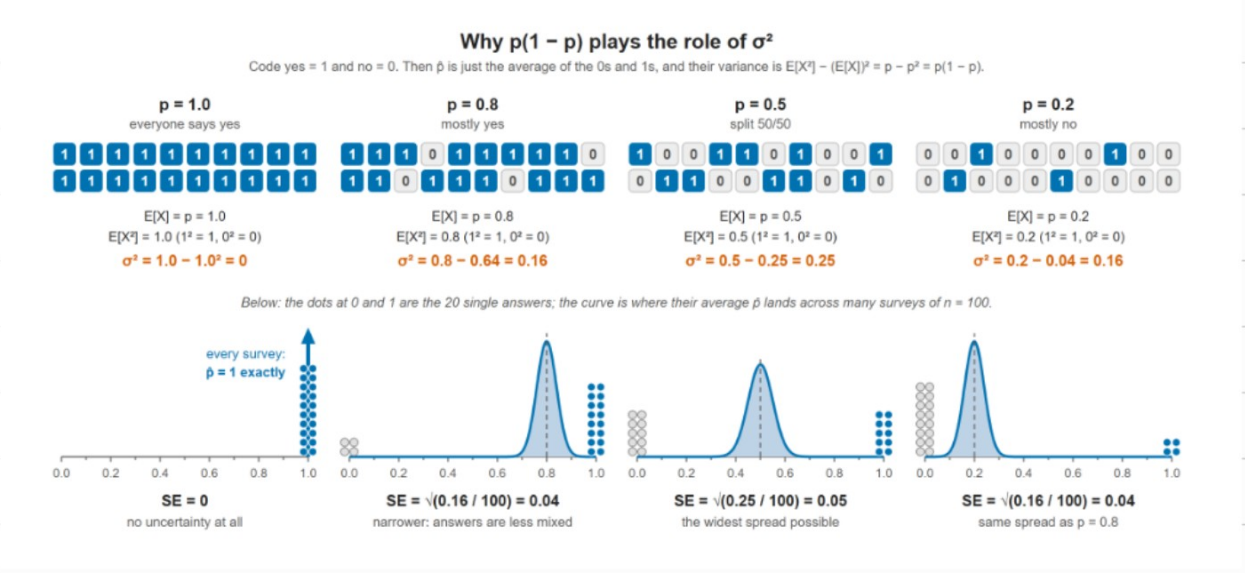
Each dot is one simulated survey. Its $\hat{p}$ misses the true $p=0.5$ by different amount each time, but about 95% of the dots land inside $\pm E$. The bottom row previews point 4: four times as many people makes the curve half as wide

3. Why $p(1-p)$ plays the role of σ^2. Code each answer as 1 (yes) or 0 (no). Then $\hat{p}$ is just the average of those 0s and 1s and their variance is

$$\sigma^2 = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

Since $1^2=1$ and $0^2=0$. Intuitively:

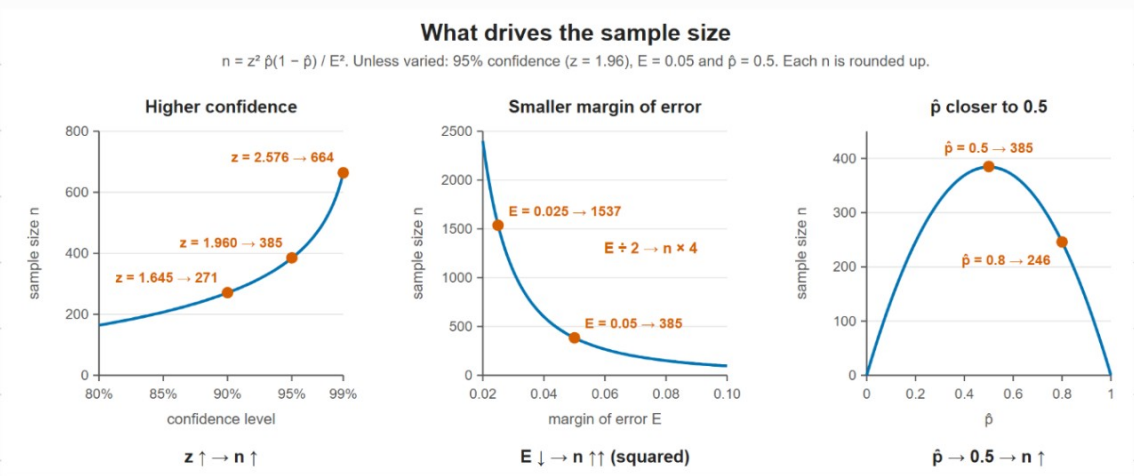
- If everyone say yes ($p=1$) every sample looks the same: $p(1-p)=0$, so there is no uncertainty.
- If the population is split 50/50, sample can easily tilt either way, so the variation is largest $p(1-p)=0.25$



A single answer can only be 0 or 1 (the dots), but the average of 100 answers clusters tightly around p (the curve). The more mixed 0s and 1s, the bigger $p(1-p)$, and the wider $\hat{p}$ spreads from survey to survey. $p=0.2$ and $p=0.8$ give the same spread, since $0.2\times0.8=0.8\times0.2$: only how mixed the answer are matters, not which side is the majority.

4. Why divide by n. In a small sample, a few unusual answers move $\hat{p}$ a lot. In large one they get diluted. The variance shrinks like $1/n$ so SE shrinks like $1/\sqrt{n}$ Because of the square root, halving the margin of errors take 4 times as many people.

5. Turn it around. Normally you collect data and then compute E. Sample-size planning runs the other way: you choose E you want and solve for the n that achieves it.



Change	Effect on n	Why
Higher confidence ($z\uparrow$)	$n\uparrow$	Being more sure needs more evidence
Smaller margin of error ($E\downarrow$)	$n\uparrow\uparrow\uparrow$ (squares)	More precision need more data
p closer to 0.5	$n\uparrow$	A more mixed population is less predictable

Each panel changes one ingredient and holds the other at 95%, $E = 0.05, \hat{p} = 0.5$. The middle panel is the steepest: halving E multiplies n by 4.

6. **Why $\hat{p} = 0.5$ is the safe choice.** You plan the survey before you have the data, so p is unknown. The safe move is to use the $\hat{p}$ that makes n as large as possible. Since z and E are fixed, that mean finding the maximum of

$$f(\hat{p}) = \hat{p}(1-\hat{p}) = \hat{p} - \hat{p}^2, \quad 0 \le \hat{p} \le 1$$

With calculus. Set the 1st derivative to zero

$$f'(\hat{p}) = 1 - 2\hat{p} = 0 \rightarrow \hat{p} = 0.5$$

Check the second derivative:

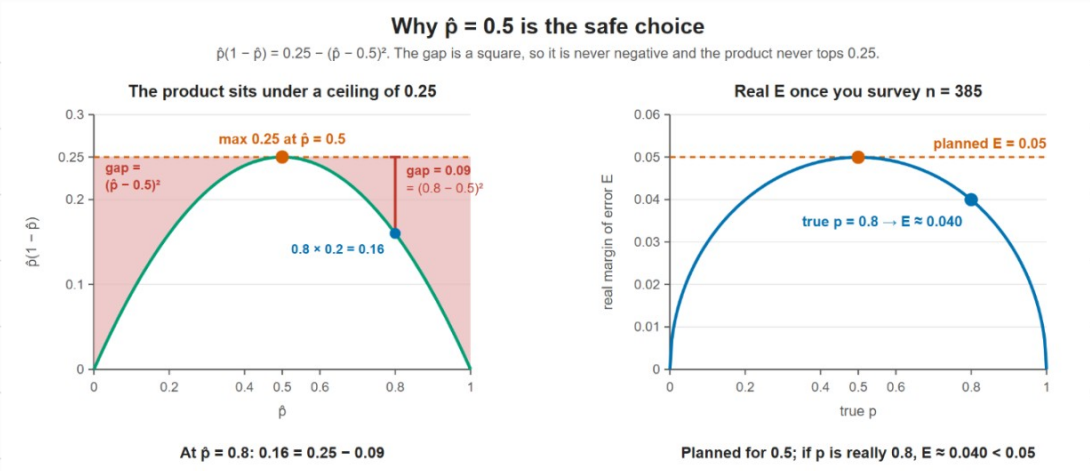
$$f''(\hat{p}) = -2 < 0 \rightarrow \text{the curve bends downward everywhere}$$

Thus the $\hat{p} = 0.5$ is the maximum, not a minimum. Its value is $f(0.5) = 0.5 \times 0.5 = 0.25$. At the ends $f(0) = f(1) = 0$, so 0.25 is the largest value on the whole range $0 \le \hat{p} \le 1$.

Without Calculus. Completing the square gives the same answer:

$$\hat{p}(1-\hat{p}) = 0.25 - (\hat{p} - 0.5)^2$$

The square term is never negative, so the product can never exceed 0.25, which happens only at $\hat{p} = 0.5$. Planning for this worst case guarantees n is large enough. If the true p turns out to be, say, 0.8, the real margin of error simply come out a bit smaller than planned.



Left: the shaded gap between the curve and 0.25 is $(\hat{p}-0.5)^2$, which is zero only at $\hat{p} = 0.5$

Right: Survey planned $n = 385$ and if p turns out to be 0.8 the margin of error is $1.96\sqrt{\frac{0.16}{385}} \approx 0.040$, comfortably inside 0.05 you asked for.

> In one Sentence: Sample answers wobble around the truth. The wobble grows with how mixed the population is and shrinks with $\sqrt{n}$

So you pick n large enough that z times the wobbles fits inside the error you'll accept.

3. Example 1 - Working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many student were sampled?

$$n = z^2 \times \frac{\sigma}{E^2} = \frac{1.96^2 15^2}{5.367^2} \approx 30 \rightarrow n = 30 \text{ student}$$

* the mean 84 isn't needed.

4. Example 2 - Planning a Survey on Proportion.

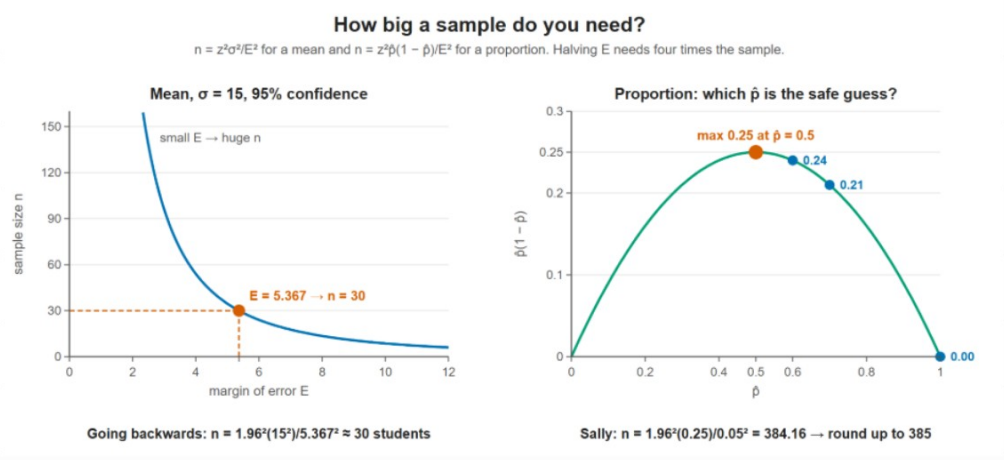
Sally wants to estimate the proportion of high-school students planning to go college. with 95% confidence and a 5% margin of error ($E = 0.05$) She has no prior estimate of $\hat{p}$.

Which $\hat{p}$ makes $\hat{p}(1-\hat{p})$ largest?

$\hat{p}$	$1-\hat{p}$	Product
1	0	0
0.7	0.3	0.21
0.6	0.4	0.24
0.5	0.5	0.25 largest

$CL = 95\% \rightarrow z = 1.96$

$$n = z \times \frac{(\hat{p}.\hat{q})}{E^2} = 1.96 \times \frac{0.5 \times 0.5}{(0.05)^2} = 384.16 \rightarrow n = 385$$



Left: the required n grows fast as the margin of error shrinks, and example 1's $E = 5.367$ corresponds to $n = 30$.

Right: $\hat{p}(1-\hat{p})$ is largest at $\hat{p} = 0.5$, so using 0.5 when nothing is known gives the safest (largest) sample size.

4. Key Takeaways

- Mean: $n = \frac{z^2 \sigma^2}{E^2}$. Proportion: $n = z^2 \frac{\hat{p}(1-\hat{p})}{E^2}$
- Both come from solving $E = z \times SE$ for n. For yes/no data the variance is $p(1-p)$, which take the places of σ^2
- With no estimate of $\hat{p}$ use 0.5. This gives the largest, safest n.
- Always round up.
- A smaller margin of error or higher confidence level requires a larger sample.

9.6 Chebyshev's Theorem

1. Statement

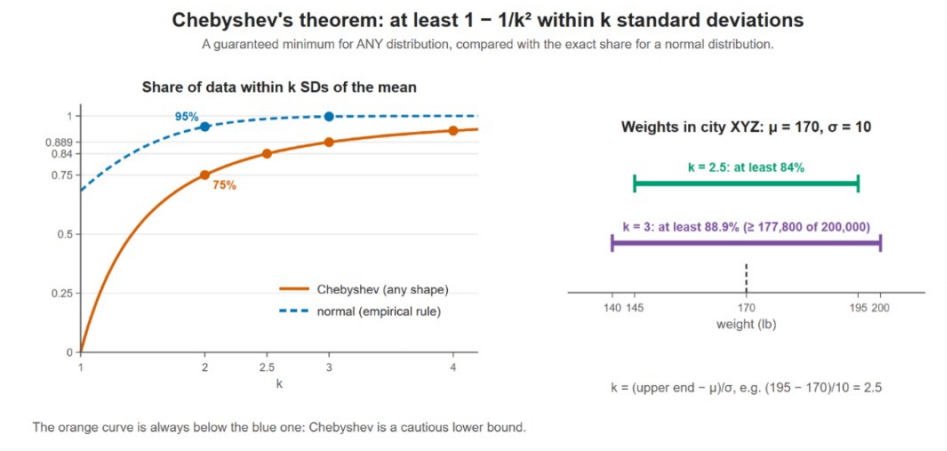
For any distribution, whatever its shape, the proportion of data within k standard deviations of the mean is at least

$$1 - \frac{1}{k^2}, \quad k > 1$$

- It gives a minimum, not an exact value. The true proportion can be higher.
- It works for all distribution, not only normal ones.

2. Chebyshev vs the empirical rule

Within	Empirical rule (normal only)	Chebyshev (any distribution, minimum)
1σ	68%	- (k must be >1)
2σ	95%	$1 - \frac{1}{4} = 75\%$
3σ	99.7%	$1 - \frac{1}{9} = 88.9\%$
4σ	$\approx 99.99\%$	$1 - \frac{1}{16} = 93.75\%$



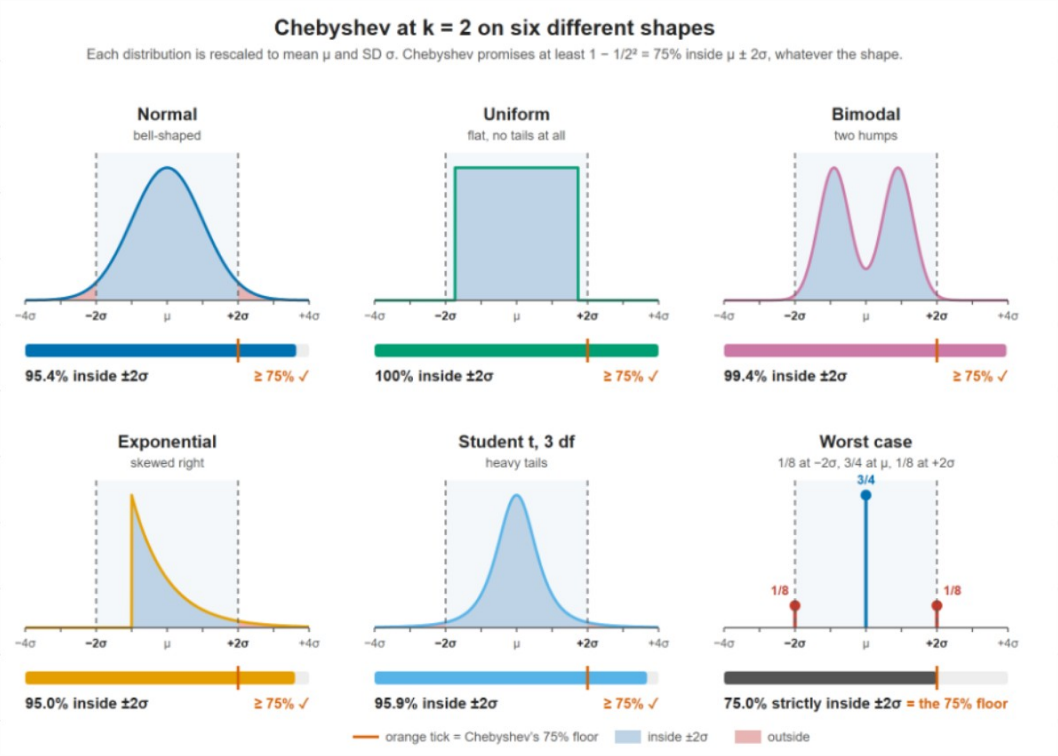
Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue).

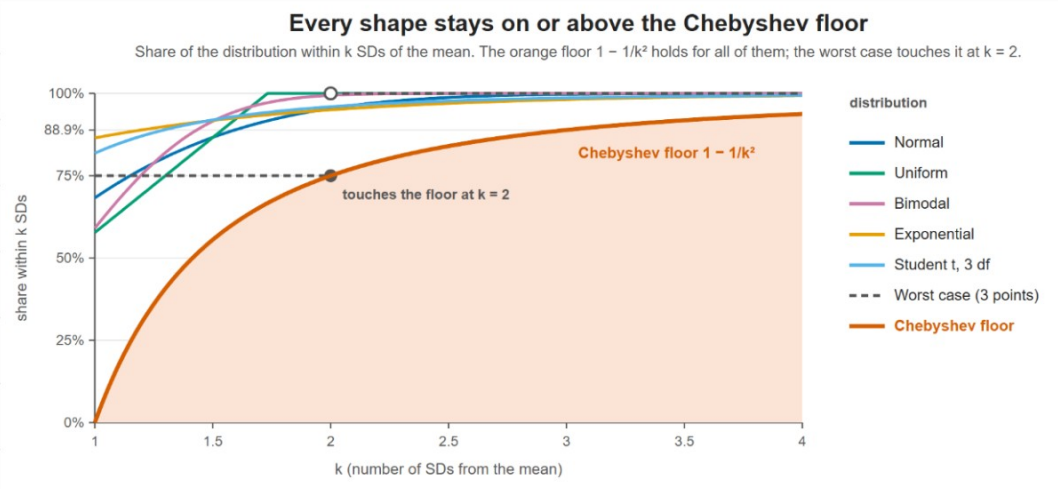
For example, it guaranteed 75% within 2 SDs, while a normal distribution has 95%.

Right: the two intervals from the weight example.

3. Chebyshev on other famous distribution

"Any distribution" is a strong claim, so here it is tested on six very different shapes. Each one is rescaled to the same mean μ and standard deviation σ, so the axis reads directly in SDs from the mean.





The orange curve is the floor $1-\frac{1}{k^2}$. Each coloured line shows the true share within k SDs for one shape, and none dips below the floor. The dashed worst case touches it at $k=2$, which is why Chebyshev's theorem is sometimes called "sharp".

What "sharp" means. A bound is sharp (or tight) when it can't be improved. Suppose someone proposes stronger rule, such as "at least 80% of any data lies within 2σ" the worst case breaks it, because it has only 75%. So 75% is the best guarantee that holds for every possible shape. Real data usually beats it easily, but no universal rule can promise more.

A worst case for any k. The example above touches the floor only at $k=2$ because it was built for $k=2$ The same recipe works for any $k>1$:

Value	$\mu - k\sigma$	μ	$\mu + k\sigma$
Share of the data	$\frac{1}{2k^2}$	$1 - \frac{1}{k^2}$	$\frac{1}{2k^2}$

- Its mean is μ. Because the two outer points balance each other
- Its SD is exactly σ. Only the outer points are away from the mean, each by $k\sigma$:

$$\text{Variance} = \frac{1}{2k^2}(k\sigma)^2 + \frac{1}{2k^2}(k\sigma)^2 = \frac{\sigma^2}{2} + \frac{\sigma^2}{2} = \sigma^2$$

- Exactly $\frac{1}{k^2}$ of the data sits $k\sigma$ from the mean so the share strictly inside $\mu \pm k\sigma$ is exactly $1 - \frac{1}{k^2}$, right on the floor.

With $k=2 \rightarrow \frac{1}{8}, \frac{3}{4}, \frac{1}{8}$ at $\mu - k\sigma$, μ, $\mu + k\sigma$

$k=3 \rightarrow \frac{1}{18}, \frac{8}{9}, \frac{1}{8}$

Each worst-case example just touches the curve at its own k.

4. Finding k

If an interval is centered on mean $\mu \pm k\sigma$ then $k = \frac{x - \mu}{\sigma}$

Using the upper end x (the lower and gives the same k).

5. Example - Weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape is unknown.

(a) Minimum Percentage of people weighing $145 < x < 195$ lb

$145 \rightarrow k = \frac{170 - 145}{10} = 2.5$

$195 \rightarrow k = \frac{195 - 170}{10} = 2.5$

* Both ends are 25 lb or $k=2.5$

$1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 0.84$

So $P(145 \leq X \leq 195) \geq 84\%$

(b) The city has 200 000 resident. What is the minimum number weighing between 140 and 200 lb?

Both ends are 30 lb from the mean, so

$k = \frac{200 - 170}{\sigma} = 3 \rightarrow 1 - \frac{1}{k^2} = 1 - \frac{1}{3^2} = \frac{8}{9} = 0.889$

$\frac{8}{9} \times 200{,}000 \approx 177{,}778$ people

So $P(140 \leq X \leq 200) \geq 0.889$ or $n(140 < x < 200) > 177\,800$

6. Key Takeaways

- Chebyshev: at least $1 - \frac{1}{k^2}$ of any set lies within k SDs of the mean ($k>1$)
- Use it when the distribution is unknown or not normal. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean
- Answer are lower bounds: say "at least"